

The Impact of Elevated Ambient Ozone Levels on Adverse Birth Outcomes Among Pregnant Women in South Korea

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December 14, 2024

Abstract

Prominent rises in ambient ozone levels are found within the East Asian region and are suspected to have a direct impact on the occurrence of low birth weight in newborns. Extensive literature also shows the negative consequences associated with air pollutants and adverse birth outcomes. These issues motivate our analysis of ozone impacts on low birth weight in Korea. In this paper, we show that the ongoing increase in ozone is associated with a statistically significant rise on the occurrence of low birth weight in newborns. We suggest that further research be conducted in this limited area of study to understand the impacts of ozone levels on adverse birth outcomes. We discuss potential policies to influence individual decision-making to reduce ozone pollution and mitigate health issues such as low birth weight in newborns.

JEL Codes: K32, K10, Q52, Q53, Q56, Q58, I14 I28, I31, J12, J13, L94.

Keywords: low birth weight; ozone; particulate matter 10; policy; EPA standard

I. Introduction

In East Asia, surface ozone concentrations have rapidly increased since the 2000s and are projected to continue rising in the coming years. Over the past two decades, South Korea has experienced a significant surge in ozone levels, consistently exceeding the World Health Organization (WHO) standard of 60 parts per billion (ppb), with levels occasionally reaching as high as 96 ppb, as recorded by AirKorea sites (Colombi et al., 2023). Despite this alarming upward trend, there has been little success in implementing effective measures to curb the ongoing rise in ozone levels. This situation raises concerns about the severe consequences of high ozone levels, particularly for vulnerable populations such as infants and the elderly. In 2016, ambient air pollution was associated with 6.5 million premature deaths globally, with almost 94% of these deaths occurring in low and middle income countries (WHO, 2018). Unborn babies, newborns, and young children are especially vulnerable to the harmful effects of air pollution due to their developmental stage, making them more susceptible to lifelong health problems. Numerous academic and medical studies have linked exposure to ambient air pollution with low birth weight (LBW) in newborns, a condition associated with an increased likelihood of adverse health outcomes later in life. These outcomes include food allergies, congenital disabilities, impaired cognitive abilities, and difficulties forming romantic relationships and becoming parents (Nosarti, 2019). Such challenges can ultimately diminish the well-being and mental health of affected individuals. In South Korea, while the total number of births has been declining over the past decade, the prevalence of low birth weight has been rising. Several factors may contribute to this troubling trend, including advanced maternal age, worsening air pollution, and the impacts of the COVID-19 pandemic.

This paper focuses on the impacts of air pollution on low birth weight and seeks to answer the question of what individuals can do to mitigate the effects of ambient ozone. The increase in ozone levels results from pollutants emitted by cars as well as by large corporations operating massive power plants, industrial boilers, refineries, and chemical plants (EPA, 2024). This scenario of large corporations affecting individuals in society can be conceptualized using a model of precaution (Shavell, 2003). Here, we assume that the injurer is a large corporation, such as the Korea Electric Power Corporation (KEPCO), which operates multiple power plants, and the victims are Korean citizens. In this unilateral model, the goal would be for KEPCO to choose a level of precaution and activity that maximizes the net social benefits of running its power plants. KEPCO would need to balance the costs of pollution imposed on Korean citizens against the benefits, such as generating electricity for the country. This model includes several assumptions - such as no opportunity for negotiation, risk neutrality for both KEPCO and Korean citizens, among others - and acknowledges additional complexities, including Korean citizens' contribution to pollution through car emissions and the difficulty of pinpointing specific health outcomes caused by environmental factors. While not a perfect comparison, this hypothetical scenario highlights the role individuals play in influencing ozone levels and how these dynamics affect broader societal constituents.

The potential impact of increased ozone pollution on adverse birth outcomes in Korea, combined with the choices made by both corporations and individuals that influence public health, motivates us to explore this important area of research. In this paper, we aim to examine the relationship between ozone levels and low birth weight in Korea, while also considering policy implications and exploring how these policies can affect decision-making among both corporations and individuals. We hypothesize that an increase in ambient ozone levels from an

average of 22.5 ppb to 32.7 ppb between 2010 and 2023 will significantly elevate the incidence of low birth weight among pregnant women in South Korea. We conduct two analyses, a primary analysis consisting of a benchmark model and an auxiliary model, both of which are two-way quadratic fixed effects models. Then we look into a secondary analysis which acts as a case study to assess the impact of higher levels of ozone pollution on the model. This secondary analysis uses a two-way fixed effects model with a dichotomous variable of interest. The findings of our study show that increases in ozone lead to a rise in the incidence of LBW in newborns. This result is further corroborated by our auxiliary model, which substitutes ozone levels with particulate matter 10 (PM10) levels, as literature indicates PM10 also impacts low birth weight. We observe a similar increase in low birth weight as PM10 levels rise. However, the results from our secondary analysis are contentious, as they indicate that unhealthy ozone levels rather than healthy ozone levels seem to reduce the number of newborns with low birth weight in Korea. This coefficient is not statistically significant, with a p-value of 0.923.

The paper is structured as follows. The literature reviewed and referenced is presented in Section II. Subsequently, Section III details the data used, while Section IV outlines our identification strategy. Building on this foundation, Section V reports the main empirical results and secondary analysis. Finally, Section VI offers concluding remarks.

II. Literature Review

Currently, there is limited research on the impact of ozone emissions in Korea on adverse health outcomes. Instead, the existing studies primarily focus on the effects of particulate matter (PM₁₀ and PM_{2.5}) on mortality rates. Notable examples include Baek, Altindag, and Mocan (2015), who examined the effects of yellow dust air pollution (known as Hwang-sa in Korean), linking it to an increased likelihood of low birth weight and premature births, while also noting an increase in gestational age. Similarly, Ahn et al. (2024) found that higher levels of aerodynamically exposed particulate matter were associated with adverse pregnancy outcomes in Korea. Suh et al. (2009) employed an extended Cox model with time-dependent PM₁₀ exposure to conclude that the effects of particulate matter exposure varied depending on the exposure period of neonates. Furthermore, research examining the impact of ambient ozone pollution on adverse health outcomes is even more limited. Most leading studies in this area have been conducted in China, particularly in Guangzhou, Beijing, and Baoming, focusing on ozone exposure before and during pregnancy. A study in the United States highlighted the potential for maternal ozone exposure to cause systemic inflammation and oxidative stress, which contributed to fetal growth restriction. Chen et al. (2023) conducted a multicity study in China that yielded plausible but inconclusive results regarding the association between ozone and low birth weight. While previous studies on air pollution and adverse birth outcomes in Korea have primarily focused on the effects of particulate matter on birth weight and preterm birth, and some international research outside of Korea has investigated the impact of ozone emissions with inconclusive results, our study aims to examine the intersection of existing scholarship.

Existing literature in Korea assessing the association between low birth weight and air pollution often includes socioeconomic covariates that directly affect these outcomes. Similarly,

our model incorporates these covariates to estimate the impact of pollution on birth outcomes. Key socioeconomic variables include parental education level, identified as the strongest predictor of low birth weight and preterm birth among five measures, parental nationality as a proxy for limited Korean language ability, parental age as an indicator of life expectancy, and unemployment status (Morgen et al., 2008). Shin et al. (2012) support the significance of socioeconomic factors in preterm birth risk, revealing that parents with educational attainment below high school faced the highest odds of preterm birth (odds ratios of 1.15 for fathers and 1.16 for mothers). While concurrent studies, such as Ahn et al. (2024), address major socioeconomic factors like parental education, age, and income, our research uniquely examines additional factors, including a nationwide ozone policy, marital status, birth order, and total fertility rate. Furthermore, our research distinguishes itself by utilizing data from the Korean Statistical Information Service (KOSIS), whereas most existing studies rely on the Korea National Statistical Office or ambulatory prenatal clinic questionnaires for socioeconomic demographic information.

In this paper, we investigate the impact of increased ambient ozone levels on outcomes after pregnancy, specifically focusing on low birth weight, using a two-way quadratic fixed effects model. Our research contrasts with the existing body of work that analyzes exposure to elevated levels of air pollution before and during pregnancy, particularly during the first, second, and third trimesters. For instance, Suh et al. (2008) conducted a study using a Cox model to assess exposure during the gestational period that was estimated from time-dependent survival analysis. Similarly, Chen et al. (2023) analyzed the effects of identifying critical exposure windows - specific periods of heightened susceptibility during pregnancy. Moreover, our empirical model distinguishes itself from current research by primarily focusing on low birth

weight. While previous studies have focused on mortality or sometimes explored related issues such as gestational diabetes mellitus or systemic inflammation, we concentrate specifically on low birth weight outcomes.

Prior studies have predominantly focused on the association between particulate matter and adverse birth outcomes, often neglecting the effects of rising ozone levels due to domestic air pollution in Korea. While some research has examined the impact of ozone emissions in China, the conclusions drawn have been plausible yet inconclusive. Additionally, studies in the United States have investigated the relationship between ambient ozone and adverse birth outcomes; however, these have been conducted in contexts where ozone levels are relatively stable and fall within safe limits compared to WHO guidelines. Our paper is among the first to explore the effects of ozone emissions in South Korea on low birth weight and preterm births. Given the alarming rise in ozone levels in South Korea and the simultaneous increase in low birth weight noted by Kim et al. (2016), our findings are particularly significant. Moreover, they may offer valuable insights applicable to other East Asian countries facing similar challenges.

III. Data

For infant health statistics, this study utilizes nationwide registry-based data on live births from 2010 to 2023 sourced from the Korean national birth registry. Comprehensive statistics on infant and maternal health - encompassing socioeconomic variables and health status - were publicly available and obtained from Korean Statistical Information Service (KOSIS). Monthly air monitoring data was also sourced from KOSIS. The air quality data provided by KOSIS are collected by local governments and submitted to the National Institute of Environmental Research in accordance with national measurement standards. The selected datasets for this

study span from 2010-2023 and comprise both annual and monthly data across all 17 cities in Korea.

The data cleaning process involved several methodologies. All datasets were initially in Korean and required translation, which was performed using the platform's online translator to ensure consistency and usability. Each demographic dataset consisted of sub-grouped categories within a single column. For example, maternal age was divided into groups such as 15–20, 20–25, 30–35, and so forth. These datasets needed to be reshaped from long to wide format and aggregated into two or fewer groupings to improve interpretability and minimize multicollinearity between variables that were similar but grouped differently. Another key step in cleaning the dataset was the utilization of collapsing datasets, implementing multiple loops, and applying conditional checks to convert string variables to numeric values. These methods were also used to calculate averages and maximums across years or among similar sub-grouped variables. Finally, 1:1 merges were performed using the unique identifier keys *Bycity* and *Year*. Most of the reshaping and cleaning tasks were completed using Stata, although a few datasets were appended in Excel.

Table 1: Variable Descriptions

Variable	Description
LBW	Low Birth Weight (LBW) is defined as a weight of less than 2,500 grams, measured as a percentage of the total population in a given city in Korea
Ozone	Air pollutant that poses a major public health concern, measured at the city level in parts per billion (ppb)
Ozone NH	The EPA defines healthy ozone levels as 0-54 ppb. Ozone NH is a dummy variable where 1 indicates cities in a given year with ozone levels above 54 ppb
PM10	Particulate Matter 10 is an air pollutant that poses a major public health concern, measured at the city level in micrograms per cubic unit ($\frac{\mu g}{m^3}$)
Ozone Policy	In 2020 the Korean government regulated 11 air pollutants. The ozone policy is a dummy variable, where 1 indicates the nationwide policy's implementation.
PM10 Policy	2018 decree changed Korea's air quality guidelines for fine dust. PM10 policy is a dummy variable, with 1 indicating the nationwide policy's implementation.
NFB	Any child born after the firstborn, measured as a percentage of the total population in a given city in Korea
LessNineteen	The mother is under 19 years of age at the time of childbirth, measured as a percentage of the total population in a given city in Korea
MoreThirtyFive	The mother is over 35 years of age at the time of childbirth, measured as a percentage of the total population in a given city in Korea
LessHS	The mother has an education level below high school, measured as a percentage of the total population in a given city in Korea
MoreHS	The mother has an education level above high school, measured as a percentage of the total population in a given city in Korea
Unmarried	The marital status of the mother is single, measured as a percentage of the total population in a given city in Korea
TFR	The total fertility rate is the number of children a woman would have if she were to live through her childbearing years, bearing children according to current age-specific fertility rates.

Descriptions taken from Korean Statistical Information Service (KOSIS) and World Health Organization (WHO).

The definitions and descriptions of all variables included in the primary and secondary analyses are outlined in Table 1. These variables include the outcome variable for all analyses, *LBW*, as well as the independent variables of interest - air pollutants: *Ozone*, *Ozone NH*, and *PM10*. Additionally, the socioeconomic and health status covariates are *NFB*, *LessNineteen*, *MoreThirtyFive*, *LessHS*, *MoreHS*, *Unmarried*, and *TFR*. Table 1, also accounts for national policies implemented in Korea in 2018 and 2020 to address air pollutants, specifically the *Ozone Policy* and *PM10 Policy*.

Table 2: Descriptive Statistics

Variable	Obs	Mean	Std. dev.	Min	Max
LBW	238	6.1200	0.9133	4.3292	8.8204
Ozone	238	28.9613	4.2340	19.1667	44.3333
PM10	238	41.8347	7.2398	25.6667	61.1667
NFB	238	48.1243	4.0194	32.7149	56.4610
LessNineteen	238	0.5270	0.2517	0.0490	1.1558
MoreThirtyFive	238	25.9501	7.0842	13.4557	44.1099
LessHS	238	1.5138	0.5445	0.4410	3.3341
MoreHS	238	72.1592	5.0098	59.7767	84.7445
Unmarried	238	2.5729	0.9282	1.1163	6.5133
TFR	238	1.1235	0.2430	0.552	1.642

The summary statistics of the variables in our model are presented in Table 2. The summary statistics for our outcome variable, *LBW*, and other socioeconomic covariates are expressed as percentages of the total population, while air pollutants and *TFR* are measured in their respective units, as detailed in Table 2. The primary analysis dataset, used for both the benchmark and auxiliary models, comprises 20 unique variables and 238 observations. The secondary analysis, used for our case study, contains 17 unique variables and 238 observations. All covariates in Table 2 demonstrate the expected means, minimums, maximums, and standard deviations characteristic of a developed nation. It is important to note that while the number of

observations may seem small, each row in the dataset represents the total percentage of newborns in the entire Korean population for a specific city and year.

Table 3: Policy Statistics and Number of Observations Affected by the Policy

Variable	Freq.	Percent	Cum. Percent
Ozone Policy	68 /238	28.57%	100.00
PM10 Policy	102/238	42.86%	100.00
Ozone NH (EPA Policy)	6/238	2.52%	100.00

A three-number summary statistics chart displays the frequency of observations affected by policy implementation in our dataset, as shown in Table 3. The *Ozone Policy* and *PM10 Policy* were government-enforced regulations in Korea. The 2018 policy tightened air quality guidelines for fine dust, impacting PM10 regulations nationwide. In our dataset, 102 observations were affected by this policy (ITA, 2023). In 2020, the Korean government enforced regulatory limits on 11 air pollutants, including O_3 , with 68 observations in our dataset affected by this policy (ITA, 2023). The final policy, *Ozone NH*, was previously defined as a variable of interest. It is a dichotomous variable representing all observations where the ozone level exceeds the Environmental Protection Agency’s (EPA) healthy standard of 54 ppb. While this U.S. standard has not been implemented nationwide in Korea, it serves as a case study in our secondary analysis.

IV. Methodology

The primary analysis consists of two components, the benchmark model and the auxiliary model that serves as a robustness check. Both models utilize a two-way quadratic fixed effects approach. This choice is justified for three reasons. First, previous literature frequently employed a Cox model to examine the impact of air pollution exposure before and during pregnancy (Suh et al., 2008; Chen et al., 2003). Second, the two-way quadratic fixed effects model is particularly

well-suited for panel data, as it allows for controlling unobserved time-invariant differences across units, such as variations between cities (e.g., Seoul vs. Jeju), as well as time-period-specific factors (e.g., the COVID-19 pandemic). Third, the benchmark and auxiliary models are distinctive in that they are quadratic models that include squared terms for ozone and PM10 levels, which capture the potentially nonlinear relationship between air pollutant exposure and birth weight. This addition reflects the hypothesis that smaller decreases in ozone or PM10 levels have a greater impact when ozone levels are high compared to when they are within a healthier range. The models in the primary analysis also integrate a nationwide policy component enforced by the Korean government, with each model having its own specific *Ozone Policy* and *PM10 Policy*. The general specification of the benchmark model is as follows:

$$LBW_{i,t} = \beta_0 + \beta_1 Ozone_{i,t} + \beta_2 Ozone^2_{i,t} + \beta_3 OzonePolicy_{i,t} + \beta_j \mathcal{X}_{i,t} + \alpha_i + \tau_t + v_{i,t}$$

The outcome variable in this analysis is the percentage of low birth weight in the total population. Our variables of interest are *Ozone* and *Ozone*² (measured in ppb). We also include a dichotomous variable for the Korean national ozone policy, implemented in 2020, where 1 represents observations after the policy was implemented, and 0 represents observations before 2020. The vector $\mathcal{X}$ consists of a range of control variables, including whether the child is the first-born, the mother's age at birth (younger than 19 or older than 35), maternal education (below or above high school), and marital status. Each of these covariates has an omitted category not to be confused with a reference category, as they are not dummy variables, which is the standard (e.g. mothers aged 19-35 when giving birth). Lastly, we control for the total fertility rate. Alpha and tau represent the city and year fixed effects, respectively, within the model, and nu represents the error term.

This model relies on several key assumptions. First, it assumes that the selected control variables are appropriate and have a direct impact on LBW. Second, the model assumes a polynomial relationship between ozone levels and LBW rather than a linear one. This assumption could lead to higher coefficient estimates and changes in significance. While it may better capture the true nature of the relationship, it also introduces the potential for bias if the relationship is not genuinely polynomial. Third, the model assumes that incorporating the ozone policy is not redundant, given that the time fixed effects already account for changes across different periods. Another assumption is that there is no measurement error in publicly available datasets, that have an incentive to falsify information for the security and reputation of the country. Additionally, we assume that the calculation of LBW is accurate. The outcome variable is computed by aggregating the total number of births in the categories of birth weights (1.5-1.9 kg, 2.0-2.4 kg, and less than 1.5 kg), then dividing this sum by the total number of births across all weight categories, and finally multiplying by 100 to obtain the percentage value. Lastly, for the economic modeling, we assumed it was appropriate to take the yearly average of monthly ozone levels in order to implement a 1:1 merge with other datasets. Therefore, the ozone levels in the benchmark model represent the average monthly ozone levels for each year.

The auxiliary model serves to enhance the robustness of our benchmark model, as prior literature suggests that PM10 exposure impacts adverse birth outcomes (Suh et al., 2009). We would expect a positive relationship between increased PM10 levels and LBW. Therefore, if the results from the auxiliary model align with those of the benchmark model, we can have greater confidence in the validity of our findings. The auxiliary model closely mirrors the benchmark model, with the key differences being the variable of interest, PM10 levels, and the type of

policy incorporated into the model. The general specification of the auxiliary model is as follows:

$$LBW_{i,t} = \beta_0 + \beta_1 PM10_{i,t} + \beta_2 PM10^2_{i,t} + \beta_3 PM10Policy_{i,t} + \beta_j \mathcal{X}_{i,t} + \alpha_i + \tau_t + v_{i,t}$$

The outcome variable remains the percentage of low birth weight in the total population. Our variables of interest are now PM10 and the quadratic term for PM10 (measured in $\mu\text{g}/\text{m}^3$). In this model, we include a dichotomous variable for the Korean national PM10 policy implemented in 2018, where 1 represents observations after the policy was enacted, and 0 represents observations before 2018. The vector $\mathcal{X}$ consists of the same set of control variables as in the benchmark model. Alpha and tau represent the city and year fixed effects within the model and nu represents the error term.

We also incorporate a secondary analysis that examines a "hypothetical scenario" involving the impact of adhering to an additional standard set by the Environmental Protection Agency in the United States. This analysis aims to assess the effectiveness of healthy ozone levels in reducing the occurrence of low birth weight, further strengthening the robustness of our benchmark model. The general specification of the secondary analysis model is as follows:

$$LBW_{i,t} = \beta_0 + \beta_1 OzoneNH_{i,t} + \beta_2 OzonePolicy_{i,t} + \beta_j \mathcal{X}_{i,t} + \alpha_i + \tau_t + v_{i,t}$$

The outcome variable in this model is low birth weight. Our variable of interest focuses on ozone levels that exceed the EPA standard for healthy ozone levels, measured in ppb and denoted as *Ozone NH*. Unlike the benchmark model in our primary analysis, there is no quadratic term in this model, as our variable of interest is a dichotomous variable: 1 represents observations with unhealthy ozone levels above 54 ppb, and 0 represents observations with healthy ozone levels at or below 54 ppb. The ozone policy remains consistent with the

benchmark model, referring to the nationwide policy implemented in 2020. The vector $\mathbf{X}$ includes the same set of control variables as in the benchmark model. Alpha and tau represent the city and year fixed effects within the model, and nu represents the error term.

The model relies on most of the same key assumptions as in the primary analysis, with the exception of the assumption regarding the polynomial relationship between O^3 and LBW. Additionally, there are specific assumptions related to our variable of interest, *Ozone NH*. When calculating unhealthy levels of ozone, instead of using the mean ozone level within a given year, we use the maximum ozone level recorded in a month for each city and year. This approach allows us to capture the impact of the highest ozone levels in South Korea from 2010 to 2023 and increases the number of observations affected by the EPA standard of greater than 54 ppb. However, it is important to note that we still have a low number of observations.

V. Results and Discussion

Our benchmark model results, presented in Table 4, indicate that an increase in ozone levels increases the occurrence of low birth weight among newborns. The analysis includes four separate regression models with varying fixed effects, and the coefficients for *Ozone* and *Ozone*² are statistically significant across all regressions. Furthermore, we applied two-way fixed effects in different regressions with varying sets of covariates and found that, once again, the coefficients for *Ozone* and *Ozone*² remain statistically significant across all iterations. These results are further detailed in the appendix, shown in Table 7.

Table 4: Two-Way Quadratic Fixed-Effects Benchmark Model Estimates

	(1)	(2)	(3)	(4)
Ozone	0.4887*** (0.0568)	0.2287** (0.0751)	0.2668*** (0.0715)	0.2506** (0.0770)
Ozone ²	-0.0081*** (0.0009)	-0.0030* (0.0012)	-0.0046*** (0.0011)	-0.0038** (0.0012)
Ozone Policy	0.3938*** (0.1011)	0.2188* (0.0927)	1.9397*** (0.3615)	2.7568*** (0.6990)
NFB	-0.0199 (0.0133)	-0.0363 (0.0193)	-0.0060 (0.0179)	0.0402 (0.0252)
LessNineteen	-1.1351*** (0.2590)	-0.9531*** (0.2576)	-0.7723** (0.2595)	-0.5060 (0.2805)
MoreThirtyFive	0.0420*** (0.0099)	0.0260* (0.0111)	0.0057 (0.0176)	0.0007 (0.0247)
LessHS	-0.0553 (0.1212)	-0.0009 (0.1128)	0.0306 (0.1080)	0.0712 (0.1106)
MoreHS	-0.0476*** (0.0099)	0.0062 (0.0181)	-0.0143 (0.0142)	-0.0028 (0.0193)
Unmarried	0.1249** (0.0360)	0.1349** (0.0450)	0.0726 (0.0496)	0.0389 (0.0556)
TFR	-0.3292 (0.3053)	-0.2777 (0.3464)	-0.0118 (0.4071)	0.8573 (0.5324)
Constant	2.8250* (1.4024)	3.0577 (1.6111)	2.8230 (1.8652)	-1.6785 (2.2884)
City Fixed Effects	No	Yes	No	Yes
Year Fixed Effects	No	No	Yes	Yes
N	238	238	238	238
R2	0.8412	0.7497	0.8772	0.6968

* p<0.05, ** p<0.01, *** p<0.001. Standard errors in parentheses. The outcome variable is the babies born with low birth weight in Korea (LBW_{it}).

There are three variables in our model that remain statistically significant across all regressions in Table 4: *Ozone*, *Ozone*², and the *Ozone Policy*. These results indicate that ozone levels do indeed appear to influence the prevalence of low birth weight in Korea, and the inclusion of the 2020 ozone policy seems to have a measurable impact on the model. In the final model, which incorporates both city and year fixed effects, the coefficient for *Ozone*² is -0.0038, and it must be interpreted in conjunction with *Ozone*'s coefficient of 0.2506. Specifically, a one

ppb increase in ozone is associated with an estimated 0.0048 percentage point increase in LBW in Korea in 2023 when the ozone level was 32.7 ppb, controlling for fixed effects. This result is statistically significant at the 0.01 alpha level. This finding suggests that current ozone levels in 2023 increase the likelihood of LBW among newborns in Korea. Additionally, when comparing the estimated effect of ozone levels in 2023 (32.7 ppb) to 2010 (22.5 ppb), there is an estimated 0.0766 percentage point difference in LBW. Interestingly, the predicted percentage point increase in LBW in 2010 is actually higher than in 2023, despite ozone levels being lower in 2010. This counterintuitive result challenges our assumption that higher ozone levels would have a greater impact on LBW. This anomaly could potentially be explained by in fact a non-polynomial relationship. An intriguing variable in Table 4 is the control variable *LessNineteen*, which represents mothers who gave birth before the age of 19. This variable is statistically significant across all regressions except for the final model. Its negative coefficient suggests that younger mothers may be less likely to have a newborn with LBW. However, it's crucial to note that mothers under 19 account for an average of only 0.52% of the population, highlighting the small sample size for this variable.

The auxiliary model results displayed in Table 5 below, further support the findings of the benchmark model. These results show that an increase in PM10 levels increases the occurrence of low birth weight in newborns in the Korean population. Similar to the benchmark model, the analysis includes four separate regression models with varying fixed effects, and the coefficients for $PM10$ and $PM10^2$ are statistically significant within our last regressions. These results bolster the findings of the benchmark model, aligning with existing literature that links particulate matter 10 exposure to adverse birth outcomes.

Table 5: Two-Way Quadratic Fixed-Effects Auxiliary Model Estimates

	(1)	(2)	(3)	(4)
PM10	-0.0299 (0.0401)	-0.0162 (0.0315)	0.0786* (0.0381)	0.0937* (0.0381)
PM10 ²	0.0001 (0.0005)	0.0000 (0.0004)	-0.0009* (0.0004)	-0.0011* (0.0004)
PM10Policy	0.1266 (0.1341)	-0.2118 (0.1100)	2.0939*** (0.4166)	3.0190*** (0.7528)
NFB	-0.0563*** (0.0148)	-0.0436* (0.0204)	-0.0167 (0.0180)	0.0433 (0.0256)
LessNineteen	-0.6511* (0.2848)	-0.8431** (0.2463)	-0.5815* (0.2629)	-0.4118 (0.2750)
MoreThirtyFive	0.0440*** (0.0103)	0.0427*** (0.0107)	0.0039 (0.0190)	0.0079 (0.0250)
LessHS	-0.1954 (0.1357)	0.0282 (0.1180)	0.0412 (0.1103)	0.0842 (0.1114)
MoreHS	-0.0704*** (0.0116)	0.0266 (0.0178)	-0.0122 (0.0162)	0.0012 (0.0196)
Unmarried	0.1525*** (0.0389)	0.2018*** (0.0457)	0.1221* (0.0484)	0.1024 (0.0517)
TFR	-0.2881 (0.3486)	-0.6250 (0.3617)	0.2977 (0.4521)	1.0489 (0.5576)
Constant	14.3379*** (1.3671)	6.517*** (1.5636)	4.5803* (2.0974)	-0.8808 (2.4709)
City Fixed Effects	No	Yes	No	Yes
Year Fixed Effects	No	No	Yes	Yes
N	238	238	238	238
R2	0.7867	0.6466	0.8364	0.7062

* p<0.05, ** p<0.01, *** p<0.001. Standard errors in parentheses. The outcome variable is the babies born with low birth weight in Korea (LBW_{it}).

Within our regression results in Table 4, , *PM10*, *PM10*², and the *PM10 Policy*, are statistically significant in our final model (column 4). However unlike *Ozone*, *PM10* is not statistically significant across every regression with varying fixed effects. This suggests that *PM10* does appear to have an impact on low birth weight, similar to the way *ozone* affects the outcome variable. Additionally, the magnitude of the coefficients is roughly comparable, around 0.17, with the coefficient for *Ozone* being slightly higher. Specifically, a one µg/m³ increase in

PM10 is associated with an estimated 0.0168 percentage point increase in low birth weight in 2023, controlling for fixed effects. This result is statistically significant at the alpha level of 0.05. The larger coefficient for *PM10* is due to the difference in the average PM10 level compared to the O^3 level in 2023, as well as the differing units in which O^3 and PM10 are measured.

For our secondary analysis, we incorporated a dichotomous variable of interest, *Ozone NH*, which indicates when EPA standards classify ozone levels as unhealthy or above 54 ppb. The results, displayed in the Appendix Table 6, show that an increase in the count of *Ozone NH* observations decreases the occurrence of low birth weight in newborns in the Korean population. However, this finding is far from statistically significant in any of the regressions, with a p-value of 0.923 in our final model with two-way fixed effects. Consequently, the results from this secondary analysis model do not provide any meaningful validation for our benchmark model.

The results in Appendix Table 6 indicate that if a city in a given year is in an unhealthy ozone level rather than a healthy ozone level, there is an estimated 0.0131 decrease in low birth weight. This finding is far from statistically significant. These unexpected results and the lack of statistical significance may stem from the limited number of observations affected by the EPA's unhealthy ozone standard, which accounts for only 6 out of 238 total observations. What's even more surprising is the small number of observations classified as having unhealthy ozone levels, especially considering the extensive news coverage and academic articles on Korea's ongoing ozone crisis, with levels frequently exceeding the WHO standard of 60 ppb (Colombi et al., 2023). This discrepancy raises the possibility of measurement error, either in the data reported by the National Institute of Environmental Research, from which KOSIS retrieves public data, or in the accuracy of media and scholarly reports.

VI. Conclusion

This study was motivated by the rise in ozone levels in South Korea, the sparse research conducted on this topic and the understanding that human behavior, such as activity levels and precautionary measures, directly influences ozone levels. We hypothesized that an increase in O^3 from 22.5 ppb in 2010 to 32.7 ppb in 2023 would result in an increase in LBW. Our findings largely supported this hypothesis, as we found that a one ppb increase in ozone was associated with an estimated 0.0048 percentage point increase in LBW in Korea in 2023. Furthermore, our auxiliary model reinforced these results, as we also observed PM10 was statistically significant in our model. However, in our secondary case study analysis, we did not find conclusive results. Our analysis regarding causality mostly satisfies the Gauss-Markov assumptions: the model is linear in parameters, based on a representative sample of the Korean population, free from exact linear relationships between covariates, and accounts for some though not all sources of endogeneity through fixed effects, robust controls supported by existing literature, and thorough robustness checks.

If the findings in this paper are to be validated, the next question or area of focus becomes: how can governments effectively mitigate ozone pollution? While policies such as the 2020 ozone policy and the 2018 PM10 policy (ITA, 2023) have been implemented, the continued increase in O^3 levels raises concerns about their effectiveness. We propose that stricter policies are needed at the city level, specifically targeting major polluting corporations like KEPCO, as well as encouraging the adoption of alternative energy sources to reduce ozone emissions. Additionally, since individuals in society also contribute to the rise in ozone levels, policies aimed at influencing personal behaviors would be highly beneficial. By revisiting Shavell's (2003) model of precaution, we can extend this framework - where the injurer is represented by

KEPCO and the victim by the Korean citizen - to a real-world context. This extension allows us to consider how policies targeting both corporate and individuals' activity and precaution levels could significantly reduce ozone pollution and mitigate its harmful effects.

As one of the few studies that have explored ozone levels on low birth weight in Korea, there is significant potential for future research to delve deeper into this topic. Some avenues for future work include investigating preterm birth as an outcome variable. Additionally, we are especially intrigued by the prospect of extending our secondary analysis case study to examine the effects of a national ozone standard as a dichotomous variable on low birth weight. This research would be even more valuable if air pollution data capturing ozone levels exceeding 60 ppb, as cited by Columbi et al. (2023), could be obtained.

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I would like to sincerely thank Professor Graham for all his help during this research project!

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VII. Appendix

Table 6: Two-Way Fixed-Effects Case Study Model Estimates

	(1)	(2)	(3)	(4)
OzoneNH	-0.1617 (0.1832)	0.0138 (0.1405)	-0.0978 (0.1337)	-0.0130 (0.1347)
OzonePolicy	0.4441*** (0.1089)	0.2789** (0.0961)	2.0599*** (0.3744)	2.7650*** (0.7146)
NFB	-0.0430** (0.0142)	-0.0442* (0.0201)	-0.0225 (0.0181)	0.0322 (0.0262)
LessNineteen	-0.5230 (0.2897)	-0.7809** (0.2466)	-0.5266* (0.2638)	-0.3408 (0.2796)
MoreThirtyFive	0.0562*** (0.0091)	0.0438*** (0.0103)	0.0029 (0.0193)	0.0120 (0.0251)
LessHS	-0.2699* (0.1367)	0.0051 (0.1179)	0.0553 (0.1115)	0.0850 (0.1135)
MoreHS	-0.0648*** (0.0112)	0.0330 (0.0178)	-0.0094 (0.0160)	0.0031 (0.0198)
Unmarried	0.1428*** (0.0399)	0.1764*** (0.0453)	0.1306** (0.0486)	0.1135* (0.0520)
TFR	-0.0512 (0.3273)	-0.0283 (0.3372)	0.3123 (0.4410)	0.9150 (0.5408)
Constant	11.6602*** (1.1447)	4.6293** (1.5839)	6.2493*** (01.6746)	1.5468 (2.1144)
City Fixed Effects	No	Yes	No	Yes
Year Fixed Effects	No	No	Yes	Yes
N	238	238	238	238
R2	0.7860	0.6661	0.8398	0.7316

* p<0.05, ** p<0.01, *** p<0.001. Standard errors in parentheses. The outcome variable is the babies born with low birth weight in Korea (LBW_{it}).

Table 7: Two-Way Quadratic Fixed-Effects Benchmark Model Estimates with different controls

	(1)	(2)	(3)	(4)
Ozone	0.2479*** (0.0602)	0.2479*** (0.0602)	0.2716*** (0.0615)	0.2969*** (0.0682)
Ozone2	-0.0037*** (0.0010)	-0.0037*** (0.0010)	-0.0041*** (0.0010)	-0.0045*** (0.0011)
Policy		2.1830*** (0.1779)	2.5524*** (0.2768)	2.0197*** (0.5588)
NFB			0.0424 (0.0244)	0.0338 (0.0251)
LessNineteen				-0.4063 (0.2673)
MoreThirtyFive				0.0141 (0.0232)
LessHS				
MoreHS				
Unmarried				
TFR				
Preterm				
Constant	1.1775 (0.9084)	1.1775 (0.9084)	-1.302 (1.6901)	-1.1319 (1.6928)
N	238	238	238	238
R2	0.7999	0.7999	0.7290	0.7854

* p<0.05, ** p<0.01, *** p<0.001. Standard errors in parentheses. The outcome variable is the babies born with low birth weight in Korea (LBW_{it}). There are city and year fixed effects during every regression.

Table 8: Extended Two-Way Quadratic Fixed-Effects Benchmark Model Estimates with different controls

	(1)	(2)	(3)
Ozone	0.2948*** (0.0698)	0.2574** (0.0771)	0.2506** (0.0770)
Ozone2	-0.0045*** (0.0011)	-0.0039** (0.0012)	-0.0038** (0.0012)
Policy	2.1710*** (0.6039)	2.1825*** (0.6035)	2.7568*** (0.6990)
NFB	0.0345 (0.0252)	0.0371 (0.0253)	0.0402 (0.0252)
LessNineteen	-0.4667 (0.2807)	-0.4940 (0.2815)	-0.5060 (0.2805)
MoreThirtyFive	0.0089 (0.0244)	0.0052 (0.0246)	0.0007 (0.0247)
LessHS	0.0758 (0.1091)	0.0553 (0.1106)	0.0712 (0.1106)
MoreHS	0.0007 (0.0187)	-0.0049 (0.0193)	-0.0028 (0.0193)
Unmarried		0.0614 (0.0540)	0.0389 (0.0556)
TFR			0.8573 (0.5324)
Constant	-1.2193 (2.0471)	-0.4276 (2.1611)	-1.6785 (2.2884)
N	238	238	238
R2	0.7783	0.7734	0.6968

* p<0.05, ** p<0.01, *** p<0.001. Standard errors in parentheses. The outcome variable is the babies born with low birth weight in Korea (LBW_{it}). There are city and year fixed effects during every regression.

Figure 1: Density of Observations by Population Size Across Cities

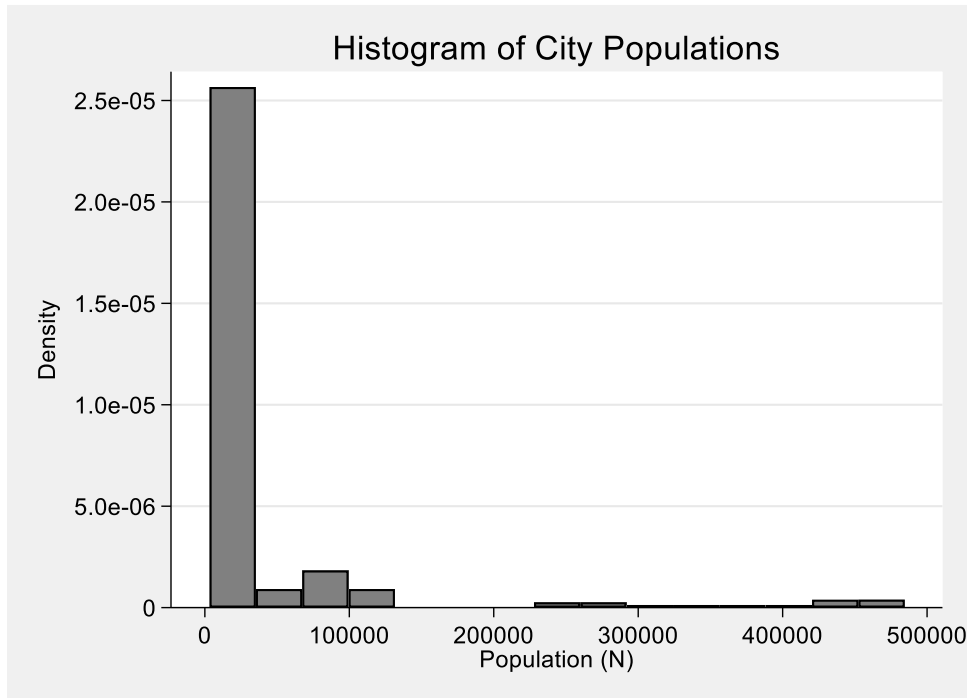


Figure 2: Density of Observations by Percentage of Low Birth Weight in Newborns

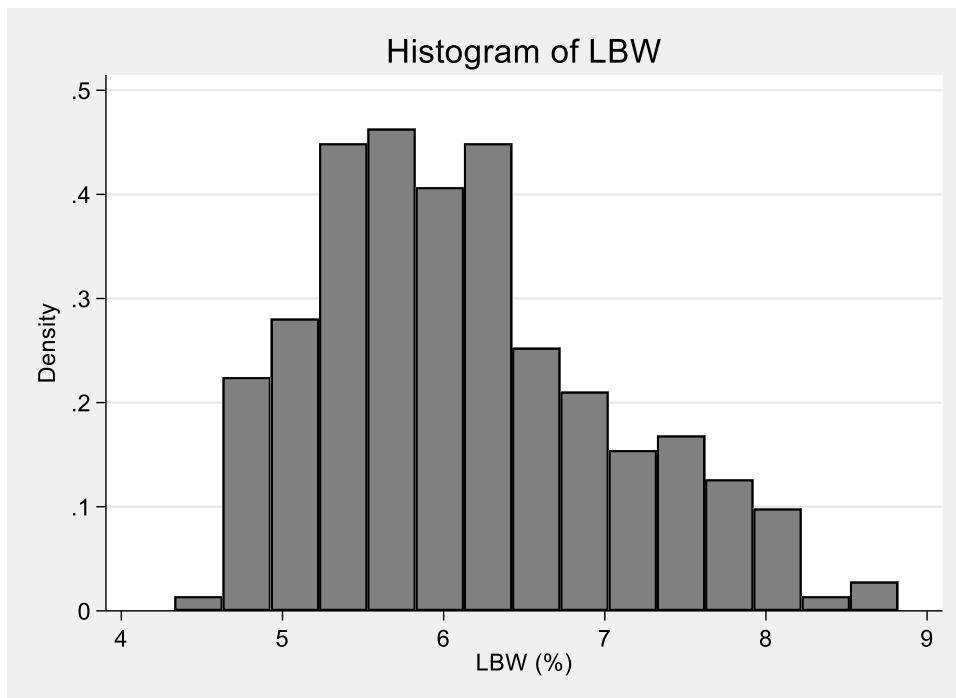


Figure 3: Density of Observations by Parts Per Billion of Ozone Levels in Primary Analysis

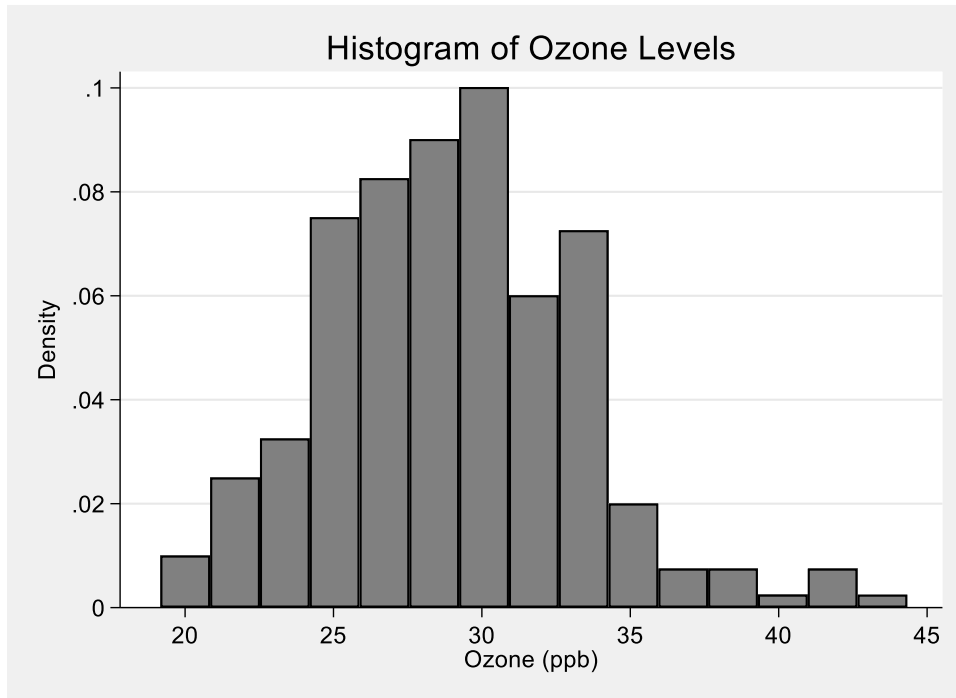


Figure 4: Density of Observations by Micrograms Per Cubic Unit of PM10 Levels

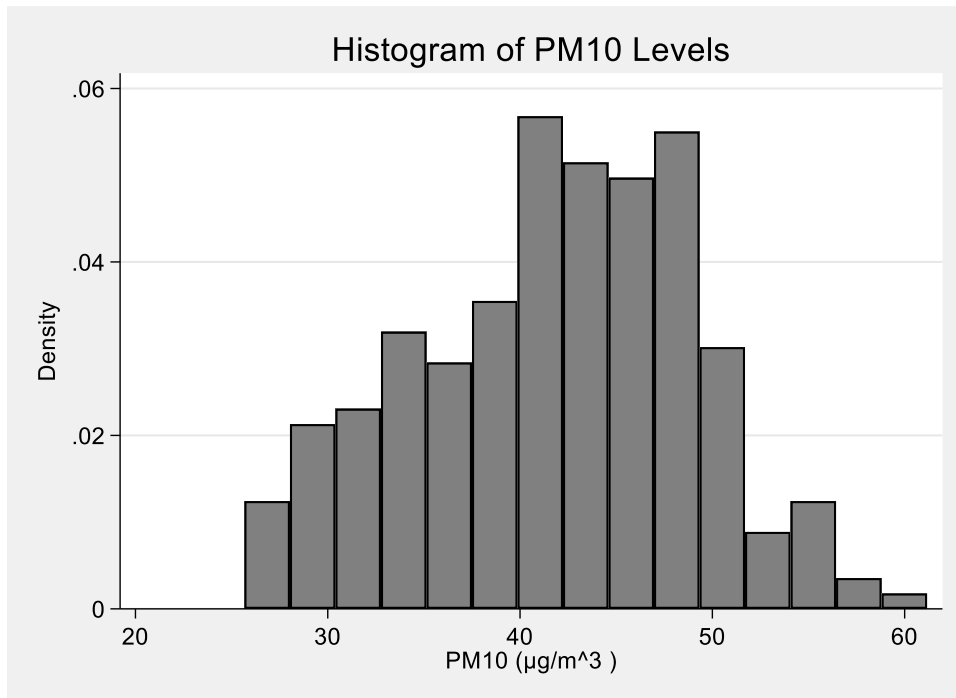


Figure 5: Density of Observations by Parts Per Billion of Maximum Ozone Levels in Case Study

